

Mst. Fahmida Sultana Naznin

Department of Computer Science and Engineering

Bangladesh University Of Engineering and Technology, Dhaka, Bangladesh

+8801740480386 | nazninfahmidasultana@gmail.com

 [LinkedIn](#) |  [github](#)

RESEARCH INTERESTS

My research domain is AI in Healthcare, with a strong focus on Natural Language Processing, Computer Vision, and Bioinformatics. I am particularly interested in integrating demographic and historical data to improve contextual insights.

WORK EXPERIENCE

• **Stamford University Bangladesh**

Lecturer in Department of Computer Science and Engineering (CSE)

Sep 2026 - Present

Dhaka, Bangladesh

- Teach undergraduate courses totaling 30 credits, developing course materials, assessments, and hands-on activities aligned with outcome-based education (OBE) objectives.
- Support student learning and academic development through project supervision, practical instruction, workshops, seminars, and curriculum enhancement.

• **Bangladesh University of Engineering and Technology (BUET)**

Research Assistant

Sep 2026 - Present

Dhaka, Bangladesh

Supervisors: [Dr. Sadia Sharmin](#), [Dr. Atif Hasan Rahman](#)

- Work under UGC-ICSETEP Research and Development Grant (RDG) project on Graph Machine Learning and Networks, focusing on GNN-LLM integration.
- Contributed to research on patient-specific clinical deterioration pathway discovery from irregular longitudinal ICU data by developing multimodal, continuous-time event graphs to identify dynamic clinical relationships and validate their necessity for mortality prediction.

• **School of Computing, National University of Singapore**

Visiting Scholar under SRI program (In person)

June 2026 - August 2026

Singapore

- Developed **ContiFuse-MM**, a continuous-time multimodal for short-term ICU mortality prediction using irregular EHR time-series and chest X-ray data from MIMIC-IV and MIMIC-CXR.
- Designed SHAP-based explainability analysis to surface important physiological features, modality contributions, and horizon-specific temporal patterns.

• **PinkLifeLine**

Founder & Chief Executive Officer

May 2024 - Present

Dhaka, Bangladesh

- Founded and lead PinkLifeLine, developing an LED-based deep learning framework for early breast cancer detection and risk prediction using longitudinal spatiotemporal imaging.
- Directed planning and execution across four flagship initiatives – Bright Beginnings, PinkTalks, PinkSpark, and Research for Hope – reaching 243 school students with sanitary product distribution and over 500,000 people community-wide.

• **Biomedical Science and Engineering Research Center (BIOSE)**

Student Researcher

Jul 2024 - Present

Dhaka, Bangladesh

Supervisors: [Dr. Md. Golam Rabiul Alam](#), [Dr. Md Rakibul Hasan](#), [Md. Mehedi Hasan Shawon](#)

- Led research on context-driven sequential transfer learning for abstractive radiology report summarization, published in ACL 2025 Findings as the first author.
- Investigated selective visual attention to high-importance regions for multimodal radiology summarization; manuscript submitted to ACL ARR in March 2026.
- Developed a foundation model for robust epilepsy detection, focusing on Asian demographic data; manuscript currently in preparation for journal submission.

• **Esolumate**

Senior Human Resources Coordinator

Oct 2022 - Feb 2024

Dhaka, Bangladesh

- Contributed to the design of an AI-based skill-sharing platform.
- Managed member onboarding, work distribution, and performance reporting.

EDUCATION

• Bangladesh University of Engineering and Technology

B.Sc, Computer Science & Engineering

◦ GPA: 3.78/4.00

2022 - Present

Dhaka, Bangladesh

PUBLICATION

C=CONFERENCE, J=JOURNAL

REFEREED CONFERENCE AND JOURNAL PUBLICATIONS

- [C.1] CBRS: Cognitive Blood Request System with Bilingual Dataset and Dual-Layer Filtering for Multi-Platform Social Streams, by **Mst. Fahmida Sultana Naznin**, Anik Saha, Zia Ul Hassan Abdullah, Anisa Binte Asad, K G Subarno Bithi, and A. B. M. Alim Al Islam. 2026. In Findings of The 64rd Annual Meeting of the Association for Computational Linguistics (**ACL 2026**). DOI:[preprint](#).
- [C.2] CSTRL: Context-Driven Sequential Transfer Learning for Abstractive Radiology Report Summarization, by **Mst. Fahmida Sultana Naznin**, Adnan Ibney Faruq, Mostafa Rifat Tazwar, Md Jobayer, Md. Mehedi Hasan Shawon, and Md Rakibul Hasan. 2025. In Findings of The 63rd Annual Meeting of the Association for Computational Linguistics (**ACL 2025**). DOI:[10.18653/v1/2025.findings-acl.1360](#).
- [C.3] MedCAR: Conflict-Aware Multi-Agent Reasoning for Multimodal Clinical AI, by **Mst. Fahmida Sultana Naznin**, Niloy Kumar Mondal, Mohammad Ninad Mahmud Nobo, Abrar Zahin Raihan, and Sheikh Azizul Hakim. 2025. Manuscript accepted at **International Conference on AI in Healthcare**.
- [C.4] Seg2Reg-Net: An Explainable AI Analysis of Predictive Limitations in Cattle Weight Estimation, by **Mst. Fahmida Sultana Naznin**, Arnab Dey, Debojit Pandit, Ahanaf Ferdous, Rakib Islam, Hasan Ibna Akbar, Jannatun Noor, and A.B.M. Alim Al Islam. 2025. In The 12th International Conference on Next-generation computing, communication, Systems and Security (**12th NSysS 2025**). DOI:[10.1145/3777555.3777559](#).

UNDER REVIEW

- [C.1] Less Is More? Selective Visual Attention to High-Importance Regions for Multimodal Radiology Summarization, by **Mst. Fahmida Sultana Naznin**, Adnan Ibney Faruq, Mushfiqur Rahman, Abrar Zahin Raihan, Niloy Kumar Mondal, Md. Mehedi Hasan Shawon, and Md Rakibul Hasan. 2025. Manuscript submitted to **ACL ARR, March 2026**. DOI:[preprint](#).
- [C.2] ContiFuse-MM: Continuous-Time Multimodal Modeling for Short-Horizon ICU Mortality in Cardiac Dysrhythmias, by **Mst. Fahmida Sultana Naznin**, and Mohammad Neamul KABIR. 2026. Manuscript submitted to **Nuerips workshops 2026**.
- [C.2] DocTalkBN: A Novel Dataset of Expert Telemedicine Conversations in Bengali, by **Mst. Fahmida Sultana Naznin**, Anik Saha, Sadatul Islam Sadi, Ananya Shahrin Promi, Wahid Al Azad Navid, and Rifat Shahriyar. 2025. Manuscript submitted to **ACL ARR, March 2026**.
- [C.3] PAN-ICA YOLO: Pyramid-Aware Neighbor-Only Inter-Channel Attention with Dense Optical Flow Mid-Fusion for Video Object Localization, by Adnan Ibney Faruq, and **Mst. Fahmida Sultana Naznin**. 2025. Manuscript submitted to **WACV 2026**.
- [J.1] BIGTNet: Anatomical Relational Reasoning for 3D Brain Tumor Segmentation, by SM Nuruzzaman Nobel, **Mst. Fahmida Sultana Naznin**, Maxine Tan, Asmaa Soliman Al-Moisheer, and Mohammad Ali Moni. 2025. Manuscript submitted to **IEEE Transactions on Medical Imaging**.
- [J.2] A Diffusion-Driven Intermediate Feature Fusion for Multi-Scale Structured Pattern Segmentation, by SM Nuruzzaman Nobel, **Mst. Fahmida Sultana Naznin**, Maxine Tan, Asmaa Soliman Al-Moisheer, and Mohammad Ali Moni. 2025. Manuscript submitted to **Artificial Intelligence In Medicine**.
- [J.3] MedSegNet-X: Hybrid Noise-Enhanced Diffusion based Medical Image Segmentation with ConvNeXt Transformer, by SM Nuruzzaman Nobel, **Mst. Fahmida Sultana Naznin**, Md All Moon Tasir, Asmaa Soliman Al-Moisheer, and Mohammad Ali Moni. 2025. Manuscript submitted to **IEEE Transactions on Multimedia**.
- [J.4] Relational Anatomical Reasoning for Anatomically Consistent 3D Brain Tumor Segmentation, by SM Nuruzzaman Nobel, **Mst. Fahmida Sultana Naznin**, Maxine Tan, and Mohammad Ali Moni. 2026. Manuscript submitted to **IEEE Transactions on Medical Imaging**.

WORKSHOPS & POSTERS

- [1] DDS-E-Sim: A Transformer-based Probabilistic Generative Framework for Simulating Error-Prone DNA Sequences for DNA Data Storage, by **Mst. Fahmida Sultana Naznin**, Swarup Sidhartho Mondol, Adnan Ibney Faruq, Debashmita Saha, Ahmed Mahir Sultan Rumi, and A. B. M. Alim Al Islam. 2025. NeurIPS 2025 Workshop SPIGM. DOI:[10.1101/2025.02.14.637785v2](#).

ONGOING WORKS

- **Political Bias Analysis** Nov 2025 - Present
Advisor: Sharifa Sultana, Assistant Professor, Dept. of Computer Science, University of Illinois Urbana-Champaign [Google Scholar](#)
 - Bias Emergence in LLM-Generated Summaries and Text Transformations from Neutral Sources in Low-Resource Settings.
- **Temporal Empathy Analysis under Uncertainty** May 2026 - Present
Advisor: Md Rakibul Hasan, Associate Lecturer (Research & Teaching) , Curtin University [Google Scholar](#)
 - Temporal-Level Multimodal Empathy Analysis under Uncertainty: Hidden Vulnerability Detection in Counseling Conversations.
- **Expert Telemedicine Conversations** Jan 2026 - Present
Advisor: Rifat Shahriyar, Professor, Dept. of Computer Science & Engineering, BUET [Google Scholar](#)
 - Built a large-scale Bengali telemedicine conversation dataset from public TV call-in shows, using LLMs to reconstruct and annotate structured doctor-patient dialogues from noisy ASR transcripts for benchmarking medical NLP and QA tasks in low-resource settings.
- **Brain Computer Interface** May 2025 - Present
Advisor: Md. Golam Rabiul Alam, Professor, Dept. of Computer Science & Engineering, BRAC University [Google Scholar](#)
 - Foundation Model for Robust Epilepsy Detection with Focus on Asian Demographic Data.
- Jan 2025 - Present
Advisor: Dr. A.B.M. Alim Al Islam, Professor, Dept. of Computer Science & Engineering, BUET [Google Scholar](#)
 - Contrastive Self-Supervised Learning with Explainable AI for Epilepsy Detection from EEG Spectrograms.
- **Longitudinal Mammogram Risk Prediction** May 2025 - Present
Advisor: Dr. A.B.M. Alim Al Islam, Professor, Dept. of Computer Science & Engineering, BUET [Google Scholar](#)
 - The More, The Better? Impact of Longitudinal Spatiotemporal Context in Mammograms for Improved Breast Cancer Risk Prediction and Tumor Localization.
- **AI-Based Self-Testing Device for Breast Cancer Patients** Oct 2024 - Present
Advisor: Dr. M Sohel Rahman, Professor, Dept. of Computer Science & Engineering, BUET [Google Scholar](#)
 - Breast Cancer Lump Detection Using Deep Learning on LED Image Data.

PROJECTS

- **Plastic4Perks: [Plastic Recycling and Gamification Platform with Carbon Tracking]** Nov 2025 - Jan 2026
Tools: [React, pygame] [\[G\]](#)
 - Developed educational games integrated with QR-based plastic collection system.
- **Speakio: [AI-powered multi-language learning platform]** Sep 2025 - Nov 2025
Tools: [React] [\[G\]](#)
 - Developed AI for lesson generation, designed software architecture, and strategized the business model.
- **QuickSolve: [AI-powered, blockchain-backed platform for complaint management]** Jul 2025 - Aug 2025
Tools: [Smart contracts, React, Node.js] [\[G\]](#)
 - Developed AI for meta data analysis, designed software architecture, and strategized the business model.
- **PinKlifeline: [AI focused breast cancer care platform]** May 2024 - Present
Tools: [Spring Boot, React, WebSocket, RabbitMq, JUnit, TestContainers, Docker, Kubernetes, Terraform, GCP, GitHub Actions] [\[G\]](#)
 - Developed AI for mammogram analysis, designed software architecture, and strategized the business model.
- **GreenBlox: [Blockchain based solution for Carbon Credit Market]** Sep 2024
Tools: [Celo, IoT, Smart contracts, React, Node.js] [\[G\]](#)
 - Developed ML model with XGBoost for solar irradiance prediction, designed algorithms for carbon reduction in solar plants, and led system architecture design.

SKILLS

- **Programming Languages:** Python, C, C++, Java
- **Web and Database:** HTML, CSS, React, Oracle, mongoDB
- **Data Science & Machine Learning:** Pandas, Numpy, Matplotlib, Seaborn, scikit-learn, PyTorch
- **Specialized Area:** Natural Language Processing, Computer Vision, Bioinformatics
- **Soft Skills:** Presentation, Problem Solving, Creative Writing, Mentoring & Collaborating

HONOURS AND AWARDS

CS competitions:

2026

- **IEEE WIE Big Idea Pitch Competition (Global)** – *Best Transformative Business Idea Pitch*
Awarded USD 1000.
- **Southeast Bank PLC. Presents Entrepren Season 3** – *1st Runner Up*
Awarded USD 1500, 1 gram Gold, and a Sylhet round-trip ticket co-sponsored by Gold Kinen, powered by Krishibid Group, driven by AIA Ventures Ltd., and in association with Noor Unique Builders.
- **MonkMoney Pty LTD Hult Prize at the University of Dhaka** – *2nd Runner Up*
Awarded USD 200 out of 311 teams, powered by HeadStart and co-powered by ExploreEd Ventures.
- **EBL Women Banking Presents Undergrad Startup Challenge 2025 Powered By AMA Coffee** – *Champion*
Awarded USD 1200; hosted by BUEDF in association with Wanpy Bangladesh.

2025

- **Blockchain Olympiad Bangladesh (Blockchain & AI Category)** – *Honorable Mention*
- **2nd ISETS-ESCAP & 3rd ISETS Global Competition of Youth Voice on Energy Transition (Global)** – *2nd prize*
Fully funded trip to UN-ESCAP Headquarters in Bangkok out of 105 projects.
- **PITCH | PROPEL | PROSPER – A Startup Showcase by SustainLaunch Labs** – *Champion*
Awarded USD 1000.
- **Idea Innovation 5.0 by Interactive Cares** – *1st Runner Up*
Awarded USD 400.
- **Huawei Seeds For The Future Bangladesh** – *Finalist (Top 21)*
- **2025 Johns Hopkins Healthcare Design Competition (Global)** – *Champion*
Awarded USD 5000 out of 440 projects from 203 universities across 44 countries, sponsored by J & J MedTech.
- **Youth Startup Summit 2025** – *2nd Runner Up*
Awarded USD 1000 out of 75,000 participants.
- **Hult Prize On Campus** – *1st Runner Up*

2024

- **ISETS-ESCAP Youth Voice Competition (Global)** – *South Asia Divisional 2nd*
Invited to Youth Dialogue at UN ESCAP, Bangkok; selected from 300 applicants across 77 teams.
- **IEEE WIE Big Idea Pitch Competition (Global)** – *Best Impact Prize*
Awarded USD 500.
- **Orange Corner Bangladesh presents IDEATION CHALLENGE 3.0** – *Champion*
Awarded BDT 100,000 out of 236 applications from 700+ participants.
- **Luminous Teen Presents THRIVE INNOVATION CHALLENGE 1.0** – *Champion*
Awarded USD 100.
- **Lean Sprint 2.0** – *2nd Runner Up*
- **Hult Prize On Campus** – *Finalist (Top 5)*
- **ISCEA Prize Global Case Competition**
Awarded USD 770 as a 70% scholarship (Top 50).
- **Blockchain Olympiad Bangladesh (AI Category)** – *Finalist (Top 16)*
- **Creaventure 3.0** – *Finalist (Top 8)*

2023

- **IEEE WIE Big Idea Pitch Competition (Global)** – *Best Impact Prize*
Awarded USD 300.
- **SympSIST: Humanitarian Project Design Idea Competition in ICCIT** – *Champion*
- **IEEE BUP Branch Fest** – *2nd Runner Up*
- **Ennovators 7.0** – *Finalist (Top-5)*
Offered an internship out of 35,000 contestants from 153 universities.

2022

- **Entrepreneurship Idea Contest & Boot-Camp, INDCON, ICCIT** – *2nd Runner Up*
- Research and Fellowships:

- 2025 [SIGHPC Immersion Program for SC25](#)
(16 Nov 2025 - 21 Nov 2025) - ACM SIGHPC Immersion Travel Grant to attend SC 2025, the International Conference on High Performance Computing, Networking, Storage, and Analysis in St. Louis, Missouri.
[BUET UIHP Cohort-5 by Research and Innovation Centre for Science and Engineering](#)
(May 2025 - August 2025) - awarded USD 300
- 2024 [Global Changemaker Fellowship Cohort-4](#) (Sep 2024 - Jun 2025) - top 33 out of 7,000 from 115 countries
[Womentor 5.0 by Banglalink](#) (Jul 2024 - Dec 2024) - top 15 STEM student out of 400+ applicants

Writing Competitions:

- 2023 National Essay Competition (Digital Bangladesh Fair-2023) - 1st;
2021 The Queen's Commonwealth Essay Competition (**Global**) - Bronze;
2020 National Human Rights Essay Competition - 4th (Dhaka Division - 1st);
2019 Eastern University Essay Competition - 13th;
2018 Talent Hunt Competition (Dhaka Divisional Best Student in Language and Literature 2018, Thana Best Student 2014-2018);
National Essay Competition on 7th March by Ideal College - 1st;
2017 International Customs Day Essay Competition - 1st;
National Children Day Essay Competition - 1st;
2016 Bangladesh Association for the Aged and Institute of Geriatric Medicine Essay Competition - 2nd;
TIB Essay Competition - Honourable Mention
2015 Essay Competition on National Mourning Day - 1st Runner Up;
National Wash Story Writing Contest - 2nd Runner Up;
2014 Essay Competition on National Energy Security - 3rd;
Nazrul Institute Essay Competition - 2nd;
Dhaka Imperial College Essay Competition - 1st;

Olympiads:

- 2019 1st LCGHS Inter School and College Fair '19 - Extempore Speech - 2nd; Earth Olympiad - 1st, Math - 3rd.
2018 Holy Cross College 17th Inter College Science Festival '18 - Chemistry Olympiad - 2nd;
2017 1st DGBHS Inter School Science Festival '17 - Chemistry Olympiad - 2nd, Extempore Speech - 1st;
10th ISTARC Science and Technology Festival '17 - Extempore Speech - 1st, Nature Olympiad - 1st;
2015 Junior Science Olympiads '15 (Regional) - 1st;

Academic Scholarships:

- 2022 Dean's Award For Academic Excellence;
2015-2016 The Scholars Forum;
2014-2015 Haji Shahab Uddin Memory Scholarship;
2013-2016 The Students Welfare Foundation;
2013, 2015 Sher-E-Bangla Memorial Scholarship;
2013 Nur MD. Sir Memory Scholarship, BICAL Foundation;
2011-2016 Bangladesh Cadet Academy.

LEADERSHIP EXPERIENCE

- | | |
|---|---|
| 1. Campus Leader , Interactive Cares | May 2025 - Sep 2025 |
| 2. Dir. of Outreach Management , Bangladeshi Women in Computer Science and Engineering | Apr 2025 - Present |
| 3. Assistant General Secretary
Membership Development Coordinator
IEEE BUET Student Branch WIE Affinity Group | Dec 2024 - Present
Jan 2024 - Dec 2024 |
| 4. Logistic Secretary
Information Secretary
BUET Debating Club | Sep 2024 - Present
Apr 2023 - Sep 2024 |
| 5. Campus Ambassador , IEEE Xtreme 18.0 | May 2024 - Mar 2025 |
| 6. Campus Ambassador , Youth Opportunities | Jan 2023 - Dec 2023 |
| 7. Campus Ambassador , AppLink | Oct 2022 - Dec 2023 |
| 8. Co-ordinating Secretary , Holy Cross College English Language Club | Aug 2019 - Apr 2020 |

VOLUNTEER EXPERIENCE

- **Volunteer** Feb 2024 - present
Agami Education Foundation 
 - Working on "[Library Project](#)" for underprivileged, distributed books for [AAlok Shikkhaloy](#), and judged [IET 2024](#).
- **Volunteer** Oct 2020 - Dec 2023
Evolution 360 
 - Created campaign content, online in charge "[Shreyasi](#)", and joined Project [YED](#) by [MJF](#).
- **Volunteer** Dec 2022 - Jan 2024
FOLV - Forum Of Liberal Volunteers 
 - Supported "[USHNOTA](#)" project, raising funds for winter clothing.

REFERENCES

1. **Dr. A.B.M Alim Al Islam**
Ph.D. in Computer Engineering, School of ECE, Purdue University, USA
Professor, Department of Computer Science & Engineering
Bangladesh University of Engineering and Technology, Bangladesh
Email: razi_bd@yahoo.com
Relationship: [Advisor, Research Supervisor]
2. **Dr. Mohammad Ali Moni**
Ph.D. in Artificial Intelligence & Data Science, University of Cambridge
Senior Research Fellow, School of Health & Rehabilitation Sciences
The University of Queensland, Australia
Email: m.moni@uq.edu.au
Relationship: [Research Supervisor]
3. **Dr. Mohammad Neamul Kabir**
Ph.D. in Computer Science, National University of Singapore
Lecturer (Educator Track), Department of Information Systems & Analytics
School of Computing, National University of Singapore, Singapore
Email: kabir.neamul@nus.edu.sg
Relationship: [Research Supervisor, NUS SRI Visiting Scholar Program]